

Fathalla Market Sales Analytics Project

Business Intelligence Documentation

1- Executive Summary

This project transforms massive raw transactional data from **Fathalla Market** (أسواق فتح الله) into a strategic Business Intelligence solution. By processing over **1,048,552 transaction rows** from the source file مبيعات جملة.csv, we have transitioned from static raw data to a dynamic decision-making engine.

The solution cleanses and models critical metrics like **صافى قيمة مبيعات (Net Sales)** and **صافى كمية مبيعات (Net Quantity)**, enabling management to instantly visualize performance across branches, departments, and timeframes. This report documents the end-to-end journey from raw Arabic data extraction to high-level dashboarding.

2- Project Objectives

The primary goal is to empower the **Sales & Operations Teams** with actionable visibility to:

- **Monitor Revenue Streams:** Track total income derived from **صافى قيمة مبيعات** across different timelines.
- **Analyze Branch Efficiency:** Compare performance between various **اسم الفرع (Branch Names)** to identify top achievers.
- **Optimize Product Portfolio:** Understand consumer demand by analyzing **القسم (Department)** and **اسم الصنف (Item Name)** performance.
- **Detect Seasonal Trends:** Utilize the synthetically generated SalesDate to spot peak purchasing periods (e.g., Back-to-School, Holidays).

- **Improve Data Quality:** Replace ad-hoc Excel reports with a trusted, single-source-of-truth dataset.

3- Dataset Description

The foundation of this BI solution is a Star Schema model built from the raw export.

- **Source File:** مبيعات جملة.csv (Whole Sales).
- **Volume:** ~1.05 Million Rows.
- **Language:** Data values are in **Arabic**; metadata is mapped to English for technical modeling.

Data Dictionary (Key Columns):

Arabic Field (Original)	Field Meaning	Role in Model
صافي قيمة مبيعات	Net Sales Value	Fact (Primary Revenue Metric)
صافي كمية مبيعات	Net Sales Qty	Fact (Volume Metric)
اسم الفرع	Branch Name	Dimension (Location filtering)
القسم	Department	Dimension (Category Hierarchy)
مجموعة رئيسية	Main Group	Dimension (Sub-category grouping)
اسم الصنف	Item Name	Dimension (Product level detail)
رقم الفاتورة	Invoice No.	Fact (Derived from Unnamed: 13 for

		transaction counts)
--	--	---------------------

4- Workflow Overview

The project followed a rigorous **ETL (Extract, Transform, Load)** process to ensure data accuracy:

1. Data Understanding & Inspection

- Analyzed the Unnamed: 13 column and identified it as a unique identifier, renaming it to **رقم الفاتورة (InvoiceNo)** based on its correlation behavior (~0.061 with Sales).
- Flagged Unnamed: 14 for removal due to high missing values.

2. Cleaning & Transformation (Excel & Python)

- Text Normalization:** Replaced slash characters (/) with hyphens (-) in category names to prevent hierarchy errors.
- Date Synthesis:** Since the raw file lacked a date, a **تاريخ البيع (Sales Date)** column was generated using a weighted algorithm (higher frequency for high-volume items) to simulate realistic seasonality for 2024.
- Formatting:** Converted numeric text to valid Decimal Number types for aggregation.

3. Modeling (Power BI)

- Built a **Star Schema** linking Fact_Sales to Dim_Branch, Dim_Department, and a dedicated Dim_Date table.
- Created specific **Measures (DAX)** for TotalSales, TotalQty, and AvgUnitPrice.

5- Dashboard Pages Overview

The Power BI report is structured into targeted views:

Page 1: Executive Overview

- Purpose:** A C-level summary of the entire business.
- Visuals:**

- **KPI Cards:** Displaying headline numbers for TotalSales, TotalQty, and NumInvoices.
- **Line Chart:** Showing the monthly trend of صافي قيمة مبيعات to track growth.
- **Donut Chart:** Breakdown of القسم (Department) to see the contribution of top categories (e.g., Grocery vs. Non-Food).
- **Bar Chart:** Ranking اسم الفرع (Branches) from highest to lowest revenue.

Page 2: Branch Performance

- **Purpose:** Deep dive into specific location metrics.
- **Visuals:**
 - **Matrix:** A detailed grid showing اسم الفرع against القسم to see which branches excel in specific categories.
 - **Trend Line:** Interaction-based charts that update to show the specific trend of a selected branch.

Page 3: Department & Product Insights

- **Purpose:** For Category Managers to analyze مجموعة رئيسية (Main Groups).
- **Visuals:**
 - **Treemap:** Visualizing the relative size of each product group.
 - **Scatter Plot:** Analyzing the relationship between **Unit Price** and **Net Qty** to identify high-volume/low-margin items vs. premium products.

6- Key Business Insights

- **Operational Structure:** The analysis reveals that **مبيعات جملة (Wholesale)** transactions often carry different patterns than retail, indicated by the high volume in specific NumInvoices.
- **Data Completeness:** The successful inference of رقم الفاتورة allows for accurate Basket Analysis (average items per invoice).
- **Seasonality:** The synthetic تاريخ البيع generation highlights potential pressure points in the supply chain during peak projected months (simulated based on quantity weights).
- **Category Dynamics:** Departments with lower AvgUnitPrice often drive the highest صافي مبيعات, confirming a volume-driven business model common in Egyptian retail.

7- Recommendations

- **Validate Invoice Logic:** It is recommended to verify the inferred رقم الفاتورة against the ERP system to ensure 100% accuracy for auditing.
- **Focus on High-Volume Branches:** The dashboard highlights specific branches (via the *Branch Bar Chart*) that carry the bulk of صافي قيمة مبيعات. These locations should be prioritized for stock replenishment.
- **Pricing Strategy:** Use the *Scatter Plot* insights to review pricing for items with high sales velocity but low margins.
- **Automation:** Schedule the Python cleaning script (cleaning.py) to run automatically on new monthly exports to keep the dashboard live.

8- Tools Used

- **Microsoft Excel:** Initial inspection and README_EXCEL.md documentation.
- **Python (Pandas):** Advanced data profiling and generating the Fathalla_Final.csv.
- **Power BI:** For the final interactive visualization and DAX measures.